

Unit 5 test answer key
Part A

Question	Answer	Mark
1	D	1 (K/U)
2	A	1(K/U)
3	C	1 (K/U)
4	B	1 (K/U)
5	C	1 (K/U)
6	B	1 (K/U)
7	C	1 (K/U)
8	A	1 (A)
9	D	1 (A)
10	A	1 (T)

Part B

True

☒ False

[1 mark]

☒ True

False

[1 mark]

True

☒ False

[1 mark]

True

☒ False

[1 mark]

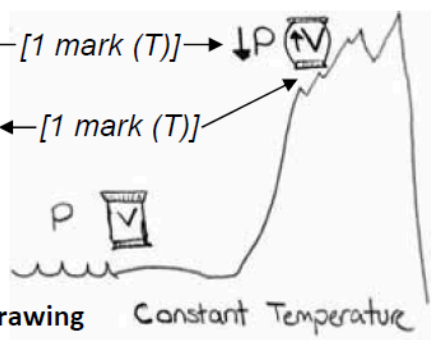
☒ True

False

Part C

1

- Pressure decreases at higher altitudes ← [1 mark (T)] →
- Volume increases as Pressure decreases ← [1 mark (T)] →



(C)] Spelling/Grammar OR Overall Logical Drawing

2

[1 mark] { G: $V_1 = 35.8 \text{ L}$ $V_2 = 1910.8 \text{ L}$ ($V_2 = 1875 \text{ L} + 35.8 \text{ L}$) $P_2 = 721 \text{ mmHg}$
 U: $P_1 = ?$

[1 mark] { E: $P_1 V_1 = P_2 V_2 \rightarrow P_1 = P_2 V_2 / V_1$
 S: $P_1 = (721 \text{ mmHg}) (1910.8 \text{ L}) / (35.8 \text{ L}) * (1 \text{ atm} / 760 \text{ mmHg})$
 $P_1 = 50.6 \text{ atm}$

[1 mark] S: ∴ the original gas pressure in the cylinder was 50.6 atm.

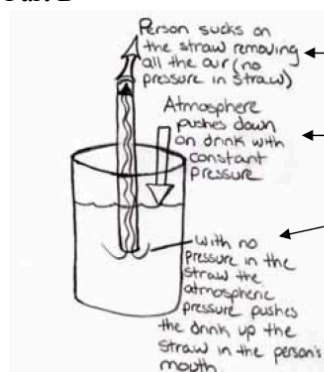
3

[2 mark] { G: $V = 12.8 \text{ L}$ $m(\text{O}_2) = 35.8 \text{ g}$ $T = 46^\circ\text{C} = 319 \text{ K}$ $M(\text{O}_2) = 32 \text{ g/mol}$
 $R = 0.082057 \text{ L atm mol}^{-1} \text{ K}^{-1}$
 U: $P = ?$

[1 mark] { E: $PV = nRT \rightarrow PV = mRT / M \rightarrow P = mRT / MV$
 S: $P = (35.8 \text{ g}) (0.082057 \text{ L atm mol}^{-1} \text{ K}^{-1}) (319 \text{ K}) / (16 \text{ g/mol}) (12.8 \text{ L})$
 $P = 2.29 \text{ atm}$

[1 mark] S: ∴ the pressure of the gas is 2.29 atm.

Part D



- ← [1 mark (T)] → - Person sucks on the straw removing all the air in the straw (no pressure)
- ← [1 mark (T)] → - Atmosphere pushes down on drink with constant pressure
- ← [1 mark (T)] → - No pressure in the straw the atmospheric pressure pushes the drink up the straw into the person's mouth

Brand of pop that travels higher.	- Diet Coke travels higher up the straw because it is less dense than coke. <i>[1 mark (K/U)]</i>
Equation for height of straw using pressure, density and gravity.	$h = \frac{P}{d \cdot g}$ <i>[1 mark (A)]</i>
Relationship between height and density at constant pressure.	- Height is inversely proportional to density <i>Or</i> - Height increases as density decreases <i>[1 mark (C)]</i>